

Thavanathan Virusan

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Jaffna, Sri Lanka

OBJECTIVE

Computer Science and Engineering undergraduate with practical experience in backend, full-stack, and distributed application development through university and personal projects. Seeking a software engineering internship to apply my programming and problem-solving skills in real-world projects.

EDUCATION

- University of Moratuwa** Mar 2024 – Present
B.Sc. (Hons) in Engineering Moratuwa, Sri Lanka
- J/Hartley College, Point Pedro** 2021 – 2023
GCE Advanced Level (Physical Science) Jaffna, Sri Lanka
3A | Z-Score: 2.3117 | District Rank: 16
- J/Hartley College, Point Pedro** 2020
GCE Ordinary Level Jaffna, Sri Lanka
9A

PROJECTS

- Second-Hand Vehicle Marketplace** [🔗]
Tools: NestJS, React, PostgreSQL, Docker, AWS
 - Designed and implemented a microservices-based marketplace for managing second-hand vehicle inventories.
 - Built API Gateway, authentication, asynchronous ETL pipeline, and search services using NestJS and PostgreSQL.
- Offline-First Emergency Communication System** [🔗]
Tools: React, Spring Boot, PostgreSQL, Google Gemini API, SMS API, Kotlin
 - Developed an offline-first disaster response system with automatic SMS fallback for emergency reporting during network outages.
 - Integrated Google Gemini API for AI-based emergency severity classification and implemented priority-based incident processing.
 - Built location-based volunteer matching using availability status and proximity to enable faster emergency response.
- Disaster Response System (J1 Device-Edge)** [🔗]
Tools: Flutter, Spring Boot, Apache Kafka, Docker
 - Developed the J1 Device-Edge module for an offline-first disaster response platform using Flutter and Spring Boot.
 - Implemented offline event queuing, automatic synchronization, and HTTP APIs for reliable emergency data submission.
 - Built a Kafka event ingestion pipeline with idempotent request handling for reliable message delivery.
- Micro Banking and Interest Management System** [🔗]
Tools: FastAPI, React, TypeScript, PostgreSQL, Docker
 - Developed a full-stack banking system for managing customers, savings accounts, joint accounts, fixed deposits, transactions, and interest calculations.
 - Implemented REST APIs with FastAPI, JWT-based authentication, role-based access control, and PostgreSQL database operations.
 - Developed automated savings and fixed deposit interest processing with scheduled background tasks and database views for reporting.
- Personal Portfolio Website** [🔗 🌐]
Tools: React, Vite, Tailwind CSS, Framer Motion
 - Designed and developed a responsive personal portfolio website to showcase projects, technical skills, education, and contact information.

SKILLS

- **Programming Languages:** Python, Java, C++, JavaScript, TypeScript
- **Frontend & Mobile:** React, Tailwind CSS, Flutter
- **Backend:** Node.js, NestJS, Spring Boot, FastAPI
- **Database Systems:** PostgreSQL, MySQL
- **Cloud Technologies:** AWS EC2, AWS S3
- **Messaging & APIs:** Apache Kafka, REST APIs, JWT
- **DevOps & Version Control:** Git, GitHub, Docker, Docker Compose
- **Languages:** English (Intermediate), Tamil (Native)

AWARDS AND CERTIFICATIONS

- Winners of the UOJ CODERS V5.0 FINALE 12hrs Hackathon | by Computer Society & the Department of Computer Science, University of Jaffna [\[🔗\]](#)
- Participated in CryptX 2.0 6hrs Competition 2026 | by Information and Communication Technology Society, University of Sri Jayewardenepura

LEADERSHIP EXPERIENCE

- **Prefect** 2021 - 2023
J/Hartley College, Point Pedro
 - Demonstrated leadership by supporting school events, maintaining discipline, and representing students while working closely with teachers and fellow prefects.

REFERENCES

Prof. D. A. Meedeniya

Professor

Department of Computer Science and Engineering

University of Moratuwa, Sri Lanka

Email: dulanim@cse.mrt.ac.lk

Dr. Adeesha Wijayasiri

Senior Lecturer

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